Vitaliy Pikalo

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EDUCATION

The Pennsylvania State University State College, PA
Bachelor of Science, Computer Engineering — Dean's List Expected Graduation: May 2028
Relevant Coursework: Data Structures & Algorithms (A-), OOP (A-), Matrices (A), Systems Programming, Computer Organization
Athletics: NCAA Division III Varsity Tennis — 17-0 in singles, AMCC All-Tournament Team

EXPERIENCE

Founder & Software Engineer — Pikalo | pikalo.org May 2026 – Present
Finance education for immigrants and first-time U.S. tax filers — Next.js, TypeScript, Supabase, Bedrock, MCP, AWS CDK

- Shipped a live platform with finance lessons, an IRS-based tax-residency checker, and FICA-refund and Roth IRA calculators; grew the waitlist to **1,300+** signups.
- Built Ask Pikalo, an AI tutor using RAG over **1,491** IRS passages that cites a source for every answer and pulls every number from calculator tools over MCP; unverifiable answers are withheld.
- Gated every push on **47** eval cases (100% pass); passed **11/11** live end-to-end tests, incl. prompt injection, at **~$0.001**/answer.

Applied AI Engineer — Penn State Dec 2025 – May 2026
OpenAI custom GPTs, prompt engineering, LLM red-teaming

- Built and deployed **2** custom GPTs used by **110+** students, restricting answers to cited sources under strict guardrails.
- Red-teamed each model with **350+** adversarial prompts from LLM agents built to force uncited, wrong, or inappropriate answers; iterated system prompts until the attacks failed before release.
- Cut student research tasks from about **2 weeks** to **minutes** while keeping a citation on every answer.

Full-Stack Software Engineer & Team Lead — Independent Contractor Apr 2025 – Aug 2025
React, Node.js, Supabase, Stripe, REST APIs

- Led a **4**-engineer team that shipped **10+** production booking apps (**100+** reservations/month); ran code review and releases.
- Built Stripe payments with Supabase Auth and role-based access control, keeping checkout PCI-compliant.

Project Ignite Engineering Lead — Carnegie Mellon University Jan 2022 – Dec 2024
TensorFlow, OpenCV, Python, PX4, thermal imaging

- Led a **12**-student team that shipped a search-and-rescue drone on an **$800** budget, donated to a Ukrainian rescue foundation.
- Built a TensorFlow/OpenCV person-detection model on the drone's thermal camera to find people in rubble, streaming their coordinates to the flight system for targeted medical-package drops.
- The system adjusted drops for wind and altitude, alerted authorities by injury severity, and texted live location updates.

PROJECTS

Levee — 3rd Place, Best Use of AWS, c0mpiled-15 | levee.rentals Oct 2026
AWS (Lambda, Bedrock), Node.js, Hono, Stripe

- Built the usage-metered API backend with per-user entitlement checks and HMAC-signed API keys, and an Amazon Bedrock feature that checks every generated figure against source data.

Kalshi Prediction-Market Data Service | GitHub 2026
Python, SQLite, pandas, scikit-learn, REST APIs

- Ran a 24/7 Linux daemon storing **270K+** market snapshots; traced an **~18x** data-inflation bug to a bad SQL join and used calibration analysis to show an apparent **+24%** return was sampling variance.

Chess960-NN — AlphaZero Reproduction | Live 2026
PyTorch, CUDA, Monte Carlo Tree Search, reinforcement learning

- Trained a **6.7M**-parameter ResNet with PUCT MCTS on **6.4M** positions plus self-play RL; reached **~1600** Elo over **100** games.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, Bash
AI & ML: RAG, LLM tool calling, MCP, LLM evals and red-teaming, Amazon Bedrock, PyTorch, TensorFlow, OpenCV
Frameworks & Data: React, Next.js, Node.js, FastAPI, PostgreSQL (Supabase), pgvector, SQLite
Open Source: 3 merged pull requests in lancedb, marimo (shipped in v0.23.12), and statsmodels
Cloud & DevOps: AWS (Lambda, Bedrock, CDK, S3), Docker, GitHub Actions CI/CD, Git, Linux